LESSON PLAN

FOLG (26)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Engine Health monitoring by using Gas path analysis
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Engine Health monitoring by using Gas path analysis
5. Specific Objectives	26.1.1.5 Trainee will be able to explain the Gas path analysis EHM Technique

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Gas Path Diagrams; Safety: Do not touch hot engine components; Instructor supervision required during diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of Engine Health Monitoring?
(b) List three common methods used for EHM techniques.
(c) Define the term 'Gas Path Analysis' in one sentence.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Gas Path Analysis • Basic principles of thermodynamic cycles • Measurement of inlet and outlet parameters • Calculation of corrected mass flow • Derivation of corrected temperature ratio	L & Ds	PPT, IB	15 min
(b)	• Interpretation of corrected flow and temperature • Identification of component efficiency trends • Detection of foreign object damage signatures • Analysis of compressor stall margins • Evaluation of turbine inlet temperature limits	L & Ds	PPT, PDF	20 min
(c)	• Comparison with baseline engine performance • Limitations of Gas Path Analysis techniques • Integration with other EHM data sources • Reporting anomalies to maintenance teams • Future trends in digital engine monitoring	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how corrected mass flow is calculated in Gas Path Analysis.
(b) Describe one limitation of using Gas Path Analysis for fault detection.
(c) How does Gas Path Analysis help in identifying compressor stall margins?

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of Gas Path Analysis learned in this lesson will assist in understanding Hot end deterioration analysis covered in LP 07.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Engine Health monitoring by using Hot end deterioration analysis	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min